

THE REMIX

LaLonde, Dehejia– Wahba, and Why the 2×2 Still Matters

The story behind yesterday's homework

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LESSON

1. LaLonde 1986

The National Supported Work Demonstration

A federally funded **randomized** job-training program.

1975–1979. Four target groups, including
disadvantaged adult men.

Random assignment to treatment vs. control. Outcome: real earnings in 1978.

LaLonde's experimental benchmark

$$\widehat{ATT}^{\text{RCT}} \approx \$886$$

1978 dollars. Adult-male subsample.

Random assignment $\Rightarrow$ no selection. This is the truth.

Then LaLonde threw away the controls

He kept the treated — the 297 NSW men.

He replaced the controls with samples from the
CPS and the **PSID**.

*A simulation of what an applied researcher would face —
no experiment, just observational data and methods.*

Then he tried every method going at the time

- OLS with controls
- Two-step Heckman selection corrections
- Fixed effects on pre-program earnings
- Difference-in-differences

Could any of them recover \$886?

No

Estimates ranged from $-\$16,000$ to $+\$3,000$.

Most were negative. Most missed the benchmark by orders of magnitude.

LaLonde (1986), AER 76(4): 604–620.

The headline that shook applied econometrics

*“The econometric procedures
cannot replicate the experimental results.”*

*This is the paper that made program evaluation
move toward randomization and design.*

LESSON

2. Dehejia & Wahba 2002

Dehejia & Wahba reopened the case

Rajeev Dehejia and **Sadek Wahba**,
Review of Economics and Statistics (2002).

They asked: does **propensity score matching**
do better than LaLonde's methods?

First, they re-cut the experimental sample

They restricted to NSW men with earnings observed in
both 1974 and 1975.

A smaller, slightly different population.

This is the subsample the homework uses: `lalonge_exp_panel.dta`.

A new experimental benchmark

$$\widehat{ATT}_{D\&W}^{RCT} \approx \$1,794$$

Roughly **\$1,700** in the way we usually report it.

Why bigger than LaLonde's \$886? Different subsample, different ATT.

Why the headlines differ

- Both numbers come from the *same* NSW experiment.
- LaLonde used the full adult-male sample.
- D&W used a *subset* — only men with two years of pre-program earnings.
- Subsetting on a covariate changes the population whose ATT you're estimating.
- Same identification (random assignment). Different target group.

This is not a contradiction — it is the ATT defined on different units.

D&W's quasi-experimental result

With **propensity-score matching** on
CPS / PSID controls...

they recovered $\approx$ \$1,700.

A constructive answer to LaLonde: design plus matching can work.

LESSON

3. What you did yesterday

Yesterday's lab was on D&W's experimental subsample

`lalonge_exp_panel.dta` — a panel of NSW men,
years 74, 75, 78.

The benchmark (post-period diff): \$1,794.

You had no covariates. Just `re`, `ever_treated`, `year`. $n_T = 185$, $n_C = 260$.

Step 1: four averages, by hand

	Pre (75)	Post (78)
Treated	$\bar{Y}_{75}^T$	$\bar{Y}_{78}^T$
Control	$\bar{Y}_{75}^C$	$\bar{Y}_{78}^C$

Four averages. Three subtractions. One DiD.

Step 2: the DiD

$$\hat{\delta}_{\text{DiD}} = (\bar{Y}_{78}^T - \bar{Y}_{75}^T) - (\bar{Y}_{78}^C - \bar{Y}_{75}^C)$$

$$= (\$6,349 - \$1,532) - (\$4,555 - \$1,267) = \$4,817 - \$3,288$$

$$\hat{\delta}_{\text{DiD}} = \$1,529$$

Post-period diff-in-means alone: $\$6,349 - \$4,555 = \$1,794$.

Two valid estimators, two different numbers

Post-period diff-in-means (78 only)	\$1,794
DiD (78 vs. 75)	\$1,529
Difference	\$ 265

The pre-period gap was $\bar{Y}_{75}^T - \bar{Y}_{75}^C = +\265 .

DiD subtracts it. Post-period diff doesn't.

Unbiasedness is a repeated-sampling concept

Random assignment makes $\mathbb{E}[\hat{\delta}] = \text{ATT}$

across hypothetical re-randomizations of the experiment.

It does *not* make $\hat{\delta} = \text{ATT}$ in any one sample.

Our pre-period happened to have +\$265 of random imbalance. Different draw, different imbalance, different DiD.

Step 3: three regressions, same number

- **OLS with interactions:** `re ~ ever_treated * post`
- **TWFE:** `feols(re ~ ever_treated:post | id + year)`
- **Long differences:** `lm(re78 - re75 ~ ever_treated)`

All three return $\hat{\delta} = \$1,529$, to machine precision.

The equivalence breaks once we add staggered timing, covariates, or weights. Tomorrow.

Step 4: the event study tells the same story

- $\beta_{74} = -\$277$ — small, close to zero. **Pre-trend check passes.**
- β_{75} is the omitted baseline (zero by construction).
- $\beta_{78} = +\$1,529$ — same as the DiD.

Each event-study coefficient is a 2×2 DiD with 75 as baseline. Same arithmetic, three points.

LESSON

4. Why the ATT didn't move

A puzzle

We used **non-experimental controls** (CPS, PSID)
in the larger LaLonde–D&W literature.

And yet the ATT we target is still $\approx \$1,700$.

Why doesn't the ATT change when we change the controls?

Because the ATT is a property of the treated

$$ATT = \mathbb{E}[Y^1 - Y^0 \mid D = 1, \text{Post}]$$

Only the $D = 1$ side appears.

The control units are just a *tool* to learn about $\mathbb{E}[Y^0 \mid D=1, \text{Post}]$.

Random assignment gave us a clean tool

Under random assignment:

$$\mathbb{E}[Y^0 \mid D=1] = \mathbb{E}[Y^0 \mid D=0]$$

So $\mathbb{E}[Y \mid D=0, \text{Post}]$ *is* the counterfactual we need.

No selection bias.

Non-experimental controls break that equality

CPS men and NSW participants are not interchangeable.

$$\mathbb{E}[Y^0 \mid D=1] \neq \mathbb{E}[Y^0 \mid D=0]$$

Selection bias enters.

NSW men had worse pre-program earnings. The CPS pool didn't.

But the *target* hasn't changed

$\mathbb{E}[Y^0 \mid D=1, \text{Post}]$ is a feature of the NSW men.

It does not depend on who we picked as controls.

What the controls control is *our estimate* of that target —
and the size of the bias.

So what is DiD for, in this story?

- In the RCT sample: DiD just *confirms* the post-period difference-in-means.
- In the non-experimental sample: DiD tries to *net out* the pre-existing level gap between NSW and CPS men.
- The truth is fixed at \$1,700. The question is whether DiD's parallel-trends assumption is enough to recover it.

Did it?

In yesterday's experimental lab: yes (in expectation)

$$\hat{\delta}_{\text{DiD}} = \$1,529 \text{ vs. benchmark } \$1,794$$

Within sampling variation. Pre-trend was flat.
Design did its job.

This afternoon, with covariates and non-experimental controls — a harder test.

LESSON

5. Now make it harder

Throw away the experimental controls

Keep the 297 NSW treated men.

Replace the controls with a random sample of
American households from the **CPS**.

Data: lalonde_nonexp_panel.dta. Same treated rows. New controls.

Selection is severe

	1974	1975	1978
NSW (treated, $n = 185$)	\$2,096	\$1,532	\$6,349
CPS (control, $n = 15,992$)	\$14,017	\$13,651	\$14,847
Diff	-\$11,921	-\$12,119	-\$8,498

$$\mathbb{E}[Y^0 \mid D=1] \ll \mathbb{E}[Y^0 \mid D=0]$$

The level gap is huge — and so is the growth-rate gap. DiD doesn't fix that.

Naive DiD on this sample

$$\hat{\delta}_{\text{naive DiD}} = +\$3,621$$

Right sign. Wrong magnitude. The benchmark is \$1,794.

NSW men had unusually low pre-earnings (\$1,532) and bounced back fast. CPS controls didn't. DiD attributes the gap in growth rates entirely to treatment.

LESSON

6. Four estimators to rescue the ATT

The roster

- **DiD with additive X** (your published spec) — one slope per covariate
- **Outcome Regression** (Heckman, Ichimura & Todd 1997) — model the trend on controls; impute the treated counterfactual
- **Inverse Propensity Weighting** (Abadie 2005) — reweight controls to look like the treated
- **Doubly Robust DiD** (Sant'Anna & Zhao 2020) — OR + IPW combined

Each lives or dies on a different assumption. Compare to the \$1,794 benchmark.

DiD with additive X

```
reg re ever_treated##post X1 X2 ... Xk, robust
```

- Pooled OLS on all years. No fixed effects.
- Covariates enter *linearly* with one slope — same pre and post.
- Identifies the ATT only if Assumptions 1–6 hold (parallel trends, no anticipation, SUTVA, homogeneous τ in X , parallel X -trends for treated and control).

This is what `lalde-did-stata.do` actually computes on the non-experimental panel.

Outcome Regression — impute the counterfactual

Use the *controls only* to model the trend:

$$\hat{\mu}_0(X) = \hat{\mathbb{E}}[\Delta Y \mid D=0, X]$$

Then impute the counterfactual change for each treated unit, and average:

$$\hat{\delta}_{\text{OR}} = \frac{1}{N_1} \sum_{i: D_i=1} (\Delta Y_i - \hat{\mu}_0(X_i))$$

Identification rests on $\hat{\mu}_0(X)$ being correctly specified.

Inverse Propensity Weighting

Estimate $\hat{p}(X) = \Pr(D=1 \mid X)$ by logit. Weight controls by:

$$w_i = \frac{\hat{p}(X_i)}{1 - \hat{p}(X_i)}$$

$$\hat{\delta}_{\text{IPW}} = \mathbb{E}[\Delta Y \mid D=1] - \mathbb{E}_w[\Delta Y \mid D=0]$$

Identification rests on $\hat{p}(X)$ being correctly specified.

Doubly Robust DiD

If *either* $\hat{\mu}_0(X)$ *or* $\hat{p}(X)$ is correct, DR is consistent.

Use the pscore as a **weight**, not as a **regressor**.

Sant'Anna & Zhao (2020). Implemented in `drdid` (Stata, R).

LESSON

7. How close did each one get?

The comparison

Estimator (Scott's published <code>drdid all</code>)	ATT	Distance from \$1,794
Naive DiD (no X)	+\$3,621	overshoots by \$1,827
DiD with additive X (your reg spec)	+\$3,522	overshoots by \$1,728
<i>Design-aware estimators (<code>drdid all</code>):</i>		
OR — Regression-adjusted (reg)	+\$1,770	−\$24
IPW — Abadie (ipw)	+\$1,861	+\$67
DR-IPW (dripw)	+\$1,979	+\$185
DR-improved (drimp)	+\$2,033	+\$239
RCT benchmark (post-period diff)	+\$1,794	—

Naive DiD and additive- X DiD both overshoot by $\sim$ \$1,800 — additive controls barely help.

The `drdid` design-aware estimators all land within \$240 of the benchmark; OR is closest at \$24 below.

OR lands closest; all four design-aware estimators are within \$240

- **OR** (reg, \$1,770) sits **\$24 below** — closest of all.
- **IPW** (Abadie, \$1,861) sits \$67 above.
- **DR-IPW** (\$1,979) sits \$185 above.
- **DR-improved** (\$2,033) sits \$239 above — furthest of the design-aware row.
- Additive- X DiD (\$3,522) barely moves off naive DiD (\$3,621).

*“Doubly robust” means robust to one misspecification, not magically best in every sample.
With $n_T = 185$ and $n_C \approx 16,000$, sampling variation alone explains the spread among design-aware estimators.*

What this lab teaches you

- The **ATT is fixed** ($\approx +\$1,794$). The controls only change the bias.
- **Additive X barely helps:** \$3,522 vs naive \$3,621. You need a design-aware estimator.
- OR, IPW, DR-IPW, DR-improved offer four different uses of the same X — all land within \$240 of the benchmark.
- On this dataset, **OR (regression-adjusted) lands closest** at \$24 below.
- No method is magic. Each rests on an assumption you can name.

The ATT lives on the **treated**.

The controls only tell you
whether your method
can **see it**.
